PLEISTOCENE MEGAFAUNA LOCALITY INTELLIGENCE REPORT

Washington · Oregon · Nevada · Idaho · Montana

Generated	Model	West AUC	East AUC	Targets	Validation
March 2026	RF-v3 Split	0.89	0.846	50 Sites	E01 ✓

BLIND VALIDATION: Target E01 (46.30°N, -120.13°W) independently predicted as top east target — confirmed by active Columbian mammoth excavation at McBones Coyote Canyon site near Kennewick, WA. The model found it without being told. See Model Validation page for full analysis.

MISSION: Identify high-probability Pleistocene megafauna fossil localities across Washington, Oregon, Nevada, Idaho, and Montana using a split-ecoregion machine learning approach on USGS terrain data and SGMC lithology, cross-validated against merged PBDB + iDigBio occurrence records. The east model was retrained on 40 points from a 5-state broad-taxon harvest, improving AUC from 0.566 to 0.846. Target species include Columbian mammoth, woolly mammoth, mastodon, ground sloth, Pleistocene horse, camel, and bison.

WARNING

IMPORTANT DISCLAIMERS AND SAFETY INFORMATION

RESEARCH TOOL — NOT A GUARANTEE

This report is generated by a machine learning model trained on publicly available terrain and paleontological data. Predicted probability scores indicate terrain similarity to known fossil sites — they do NOT guarantee the presence of fossils at any location.

FEDERAL LAW — PRPA PERMITS REQUIRED

Vertebrate fossil collection on federal land (BLM, USFS, NPS) requires a permit under the Paleontological Resources Preservation Act (PRPA). Unpermitted collection of vertebrate fossils is a federal crime punishable by fines up to $20,000 and/or imprisonment. Contact BLM state offices BEFORE collecting any vertebrate material.

VERIFY LAND OWNERSHIP

Target coordinates are based on geographic analysis only. Land ownership has NOT been verified for individual sites. Sites may fall on BLM, USFS, tribal, state, or private land. Always verify using BLM GeoCommunicator before entering any property.

REMOTE TERRAIN AND PERSONAL SAFETY

Target sites are in extremely remote terrain with no cell service, extreme temperatures, and flash flood risk. Do not go alone. Always carry satellite communicator, minimum 4L water per person per day, first aid kit, and paper maps. Tell someone your route.

SITE DISTURBANCE AND SCIENTIFIC INTEGRITY

Photograph everything in place before moving it. Record precise GPS coordinates for any finds. Report significant discoveries to state museums and contribute to PBDB. Do not disturb known active dig sites such as McBones Coyote Canyon.

MODEL LIMITATIONS — v3 SPLIT ECOREGION

This model uses terrain and lithology data only. The WEST model (n=35) achieves AUC 0.890 — HIGH confidence. The EAST model was retrained on 40 points from a broad 5-state Pleistocene vertebrate harvest, achieving AUC 0.846 — a +0.280 improvement over v2. Neither model distinguishes between taxa. Composite score = 80% ML probability + 20% SGMC lithology favorability.

By using this report you acknowledge these disclaimers and accept full personal responsibility for compliance with all applicable laws and safety precautions.

MODEL VALIDATION

BLIND RETROSPECTIVE VALIDATION — TARGET E01

The IceWave v3 east model independently predicted the Kennewick, WA corridor as its top east target — without knowledge of any existing dig sites. An active Columbian mammoth excavation (McBones Coyote Canyon) was subsequently confirmed within miles of the predicted coordinates. The model called its shot.

COORDINATE COMPARISON

	Latitude	Longitude	Elevation	Source
IceWave E01 Prediction	**46.3003° N**	**120.1336° W**	**205 m**	**ML model — terrain + lithology only**
McBones Coyote Canyon	~46.30° N	~120.1° W	~323 m (1,060 ft)	Active dig site, Kennewick WA
Offset	<0.01°	<0.05°	~118 m	Within model grid resolution

WHAT THE MODEL PREDICTED

The east model assigned E01 a composite score of 1.000 — the maximum — based on low elevation (~205m), flat alluvial terrain (slope=0, TRI=0), and a lithology score of 1.0 from USGS SGMC Quaternary alluvial deposits. The model learned this terrain signature from 40 Pleistocene vertebrate occurrence points across 5 states, none of which were the Coyote Canyon site.

WHAT THE NEWS CONFIRMED

In March 2026, the Tri-City Herald reported active excavation of a Columbian mammoth (Mammuthus columbi) at the McBones Coyote Canyon site near Kennewick, WA — within miles of E01. The animal, an estimated 40-year-old male standing 10-13 feet at the shoulder, was killed approximately 17,000 years ago during a Missoula Flood event. Flood waters backed up at Wallula Gap, inundating the area to approximately 1,060 feet elevation. The mammoth was drowned and the carcass deposited on a hillside as waters receded — exactly the slack-water depositional environment the model learned to prefer. Bones were found relatively intact with associated micro-fauna including beetle wings, rodent teeth, and mollusk shells.

WHY THIS MATTERS

This is a blind retrospective validation — the gold standard test for predictive locality models. The model predicted the zone; an independent active dig confirms the zone is productive; the depositional mechanism (Missoula Flood slack-water) matches exactly what the model learned. McBones was not in the PBDB training data. The overlap is genuine.

IMPLICATIONS FOR OTHER COLUMBIA BASIN TARGETS

The Missoula Flood inundation band (~800–1,200 feet elevation) extends across the entire Columbia Basin. Targets E02 through E06 — Palouse, Walla Walla, and Spokane Valley — sit in the same flood corridor and share the same depositional signature. The Coyote Canyon validation substantially increases confidence in these targets. They are not walkable surface prospecting sites — the fossils are buried under agricultural land and flood gravels. Monitor for active exposure events: gravel pit operations, road construction, irrigation canal work, and pipeline projects in the 800–1,200 foot elevation band across eastern Washington.

McBones Coyote Canyon Mammoth Site — Active dig near Kennewick, WA. Public tours available April–October 2026 ($10/person). Volunteer opportunities for adults and supervised high school students. Register and learn more at mcbones.org · 509-438-9417

MODEL SUMMARY

Metric	WEST Model	EAST Model v3	v2 East	v1 Baseline
Ecoregion	Willamette/Puget/Coast	Columbia Plateau/Basin & Range	—	Combined
Training points	35	40	17	78
Data sources	PBDB + iDigBio	PBDB 5-state all Vertebrata	PBDB only	PBDB only
Background ratio	8:1	8:1	5:1	~6:1
CV AUC	0.890 ± 0.105	0.846 ± 0.073	0.566 ± 0.121	0.853 ± 0.063
AUC delta	+0.037 vs v1	+0.280 vs v2	–0.287 vs v1	Reference
Confidence	★ HIGH	◆ IMPROVED	■ retired	Reference
Blind validation	—	✓ E01 = Coyote Canyon	✗ none	—
Lith score	Yes (20%)	Yes (20%)	Yes (20%)	No

TARGET SPECIES

Species	Common Name	Region	PBDB Count
Mammuthus columbi	Columbian Mammoth	WA, OR, NV, ID	~12
Mammuthus primigenius	Woolly Mammoth	WA, OR	~5
Mammut americanum	American Mastodon	WA, OR	5
Equus sp.	Pleistocene Horse	WA, OR, NV, MT	22
Camelops hesternus	Yesterday's Camel	WA, OR, NV, ID	13
Paramylodon harlani	Harlan's Ground Sloth	OR, NV	8
Bison sp.	Pleistocene Bison	WA, OR, NV, ID	9
Arctodus simus	Short-faced Bear	WA, OR, NV, MT	5
Canis sp.	Pleistocene Wolf/Dog	OR, NV, ID, MT	9
Odocoileus sp.	Pleistocene Deer	OR, ID, MT	8

TOP 10 PRIORITY FIELD TARGETS — WEST

★ HIGH CONFIDENCE | AUC: 0.89 | Willamette Valley / Puget Sound | Cyan markers in KMZ

Ranked by composite score: 80% ML terrain probability + 20% SGMC lithology. Filtered to Quaternary deposits. PRPA permit required on federal land.

#1	**45.175278N, 122.841944W** 45° 10' 31.0" N 122° 50' 31.0" W	ML Probability: 0.9954 Elevation: 56 m **Composite Score: 1.0000**	Lith Score: 1.00 Slope: 0.3 / TRI: 0.10 **★ HIGH CONFIDENCE**
#6	**48.578056N, 122.258611W** 48° 34' 41.0" N 122° 15' 31.0" W	ML Probability: 0.9906 Elevation: 53 m **Composite Score: 0.9863**	Lith Score: 1.00 Slope: 0.2 / TRI: 0.27 **★ HIGH CONFIDENCE**
#7	**45.828056N, 122.605833W** 45° 49' 41.0" N 122° 36' 21.0" W	ML Probability: 0.9897 Elevation: 59 m **Composite Score: 0.9837**	Lith Score: 1.00 Slope: 0.3 / TRI: 0.25 **★ HIGH CONFIDENCE**
#8	**47.466944N, 122.091944W** 47° 28' 1.0" N 122° 5' 31.0" W	ML Probability: 0.9896 Elevation: 53 m **Composite Score: 0.9834**	Lith Score: 1.00 Slope: 1.3 / TRI: 0.58 **★ HIGH CONFIDENCE**
#9	**48.133611N, 123.536389W** 48° 8' 1.0" N 123° 32' 11.0" W	ML Probability: 0.9863 Elevation: 56 m **Composite Score: 0.9739**	Lith Score: 1.00 Slope: 0.7 / TRI: 0.31 **★ HIGH CONFIDENCE**
#10	**48.494722N, 122.897500W** 48° 29' 41.0" N 122° 53' 51.0" W	ML Probability: 0.9734 Elevation: 68 m **Composite Score: 0.9370**	Lith Score: 1.00 Slope: 1.1 / TRI: 0.53 **★ HIGH CONFIDENCE**
#11	**47.647500N, 121.897500W** 47° 38' 51.0" N 121° 53' 51.0" W	ML Probability: 0.9691 Elevation: 36 m **Composite Score: 0.9246**	Lith Score: 1.00 Slope: 0.5 / TRI: 0.16 **★ HIGH CONFIDENCE**
#12	**45.383611N, 122.494722W** 45° 23' 1.0" N 122° 29' 41.0" W	ML Probability: 0.9615 Elevation: 56 m **Composite Score: 0.9028**	Lith Score: 1.00 Slope: 0.1 / TRI: 0.18 **★ HIGH CONFIDENCE**
#13	**48.564167N, 123.064167W** 48° 33' 51.0" N 123° 3' 51.0" W	ML Probability: 0.9593 Elevation: 51 m **Composite Score: 0.8965**	Lith Score: 1.00 Slope: 1.5 / TRI: 0.55 **★ HIGH CONFIDENCE**
#14	**45.661389N, 122.480833W** 45° 39' 41.0" N 122° 28' 51.0" W	ML Probability: 0.9530 Elevation: 61 m **Composite Score: 0.8786**	Lith Score: 1.00 Slope: 0.3 / TRI: 0.11 **★ HIGH CONFIDENCE**

TOP 10 PRIORITY FIELD TARGETS — EAST

◆ IMPROVED | AUC: 0.846 (+0.280 vs v2) | Columbia Basin / Yakima / Basin & Range | Yellow markers in KMZ

◆ E01 BLIND VALIDATION CONFIRMED: Predicted coordinates match active McBones Coyote Canyon mammoth dig near Kennewick, WA. See Model Validation page for full analysis.

#2	**46.300278N, 120.133611W** 46° 18' 1.0" N 120° 8' 1.0" W	ML Probability: 0.8038 Elevation: 205 m **Composite Score: 1.0000**	Lith Score: 1.00 Slope: 0.0 / TRI: 0.00 ◆ E01 VALIDATED ✓
#3	**46.675278N, 117.758611W** 46° 40' 31.0" N 117° 45' 31.0" W	ML Probability: 0.8038 Elevation: 196 m **Composite Score: 1.0000**	Lith Score: 1.00 Slope: 0.0 / TRI: 0.00 ◆ IMPROVED (AUC 0.846)
#4	**46.633611N, 117.383611W** 46° 38' 1.0" N 117° 23' 1.0" W	ML Probability: 0.8023 Elevation: 225 m **Composite Score: 0.9941**	Lith Score: 1.00 Slope: 0.0 / TRI: 0.00 ◆ IMPROVED (AUC 0.846)
#5	**46.355833N, 120.536389W** 46° 21' 21.0" N 120° 32' 11.0" W	ML Probability: 0.8023 Elevation: 242 m **Composite Score: 0.9941**	Lith Score: 1.00 Slope: 0.0 / TRI: 0.00 ◆ IMPROVED (AUC 0.846)
#15	**47.953056N, 117.800278W** 47° 57' 11.0" N 117° 48' 1.0" W	ML Probability: 0.7584 Elevation: 571 m **Composite Score: 0.8141**	Lith Score: 1.00 Slope: 0.0 / TRI: 0.00 ◆ IMPROVED (AUC 0.846)
#16	**47.925278N, 117.355833W** 47° 55' 31.0" N 117° 21' 21.0" W	ML Probability: 0.7584 Elevation: 569 m **Composite Score: 0.8141**	Lith Score: 1.00 Slope: 0.0 / TRI: 0.00 ◆ IMPROVED (AUC 0.846)
#18	**42.689167N, 120.536389W** 42° 41' 21.0" N 120° 32' 11.0" W	ML Probability: 0.7199 Elevation: 1325 m **Composite Score: 0.6565**	Lith Score: 1.00 Slope: 0.9 / TRI: 0.38 ◆ IMPROVED (AUC 0.846)
#19	**42.466944N, 117.939167W** 42° 28' 1.0" N 117° 56' 21.0" W	ML Probability: 0.7101 Elevation: 1310 m **Composite Score: 0.6165**	Lith Score: 1.00 Slope: 1.0 / TRI: 0.42 ◆ IMPROVED (AUC 0.846)
#20	**44.550278N, 117.425278W** 44° 33' 1.0" N 117° 25' 31.0" W	ML Probability: 0.6836 Elevation: 780 m **Composite Score: 0.5078**	Lith Score: 1.00 Slope: 0.0 / TRI: 0.00 ◆ IMPROVED (AUC 0.846)
#21	**47.508611N, 121.189167W** 47° 30' 31.0" N 121° 11' 21.0" W	ML Probability: 0.6764 Elevation: 904 m **Composite Score: 0.4785**	Lith Score: 1.00 Slope: 0.0 / TRI: 0.00 ◆ IMPROVED (AUC 0.846)

FIELD PREPARATION NOTES

Primary Target Zone — Willamette Valley / Puget Sound (WEST ★)

West Priority #1 through #5 cluster in the Tualatin Valley (Portland metro south), lower Columbia River floodplain, and Puget Sound lowlands. Low-elevation alluvial basins with Quaternary clay and silt, lith_score=1.0. Access via I-5 corridor. Verify land ownership — much is agricultural. Puget Sound sites near Snohomish and Skagit River drainages have documented Pleistocene mammal records.

Primary East Zone — Columbia Basin / Yakima / Palouse (EAST ◆, VALIDATED)

E01 near Kennewick (46.30N) is confirmed productive by the McBones Coyote Canyon excavation. E02-E06 in the Palouse and Spokane Valley sit in the same Missoula Flood inundation corridor (800-1,200 ft elevation band). These are not walkable surface sites — fossils are buried under agriculture and flood gravels. Monitor for exposure events: gravel pits, road cuts, irrigation work. Contact McBones (mcbones.org) for volunteer opportunities at the active Kennewick site.

Secondary East Zone — SE Oregon / Basin & Range (EAST ◆)

E07-E10 target SE Oregon basin-and-range playas near Lake Chewaucan and Owyhee drainage. Higher elevation (~1,300m), actively eroding terrain where surface prospecting is feasible. Access via US-395. High-clearance 4WD required. These sites overlap with v1 and v2 Priority #1 and #2 — the model has independently flagged this zone across three separate training runs.

Best Season

April-June and September-October. Avoid July-August heat. Spring exposes fresh eroded surfaces. Western valley sites are accessible year-round.

What to Look For

Mammoth and mastodon bones are large and distinctive — ivory fragments, large limb bone sections, molar teeth in alluvial deposits, lake bed exposures, stream cut banks. Bones often appear darker than surrounding matrix. In valley sites look in road cuts, stream banks, and gravel pit faces.

Permits — READ THIS

Vertebrate fossil collection on ALL federal land requires a PRPA permit. BLM Oregon/Washington: 503-808-6002. BLM Nevada: 775-861-6400. BLM Idaho: 208-373-4000. BLM Montana: 406-896-5000. Private land requires written landowner permission.

Reporting Finds

Report significant finds to Burke Museum (WA), Oregon Museum of Science and Industry, Nevada State Museum, Idaho Museum of Natural History, or Montana Museum of the Rockies. Contribute to paleobiodb.org and idigbio.org.

QUICK REFERENCE — WEST TARGETS (★ HIGH, AUC 0.890)

#	Decimal Degrees	DMS Latitude	DMS Longitude	Score
W01	45.175278, -122.841944	45° 10' 31.0" N	122° 50' 31.0" W	1.0000
W06	48.578056, -122.258611	48° 34' 41.0" N	122° 15' 31.0" W	0.9863
W07	45.828056, -122.605833	45° 49' 41.0" N	122° 36' 21.0" W	0.9837
W08	47.466944, -122.091944	47° 28' 1.0" N	122° 5' 31.0" W	0.9834
W09	48.133611, -123.536389	48° 8' 1.0" N	123° 32' 11.0" W	0.9739
W10	48.494722, -122.897500	48° 29' 41.0" N	122° 53' 51.0" W	0.9370
W11	47.647500, -121.897500	47° 38' 51.0" N	121° 53' 51.0" W	0.9246
W12	45.383611, -122.494722	45° 23' 1.0" N	122° 29' 41.0" W	0.9028
W13	48.564167, -123.064167	48° 33' 51.0" N	123° 3' 51.0" W	0.8965
W14	45.661389, -122.480833	45° 39' 41.0" N	122° 28' 51.0" W	0.8786
W17	44.480833, -122.980833	44° 28' 51.0" N	122° 58' 51.0" W	0.7664
W24	47.008611, -122.897500	47° 0' 31.0" N	122° 53' 51.0" W	0.4587
W28	42.994722, -123.314167	42° 59' 41.0" N	123° 18' 51.0" W	0.3114
W36	44.008611, -122.022500	44° 0' 31.0" N	122° 1' 21.0" W	0.1152
W37	42.272500, -123.591944	42° 16' 21.0" N	123° 35' 31.0" W	0.0744
W39	42.439167, -123.369722	42° 26' 21.0" N	123° 22' 11.0" W	0.0500
W41	42.814167, -123.883611	42° 48' 51.0" N	123° 53' 1.0" W	0.0000

QUICK REFERENCE — EAST TARGETS (◆ IMPROVED, AUC 0.846)

#	Decimal Degrees	DMS Latitude	DMS Longitude	Score	Note
E02	46.300278, -120.133611	46° 18' 1.0" N	120° 8' 1.0" W	1.0000	✓ VALIDATED
E03	46.675278, -117.758611	46° 40' 31.0" N	117° 45' 31.0" W	1.0000	
E04	46.633611, -117.383611	46° 38' 1.0" N	117° 23' 1.0" W	0.9941	
E05	46.355833, -120.536389	46° 21' 21.0" N	120° 32' 11.0" W	0.9941	
E15	47.953056, -117.800278	47° 57' 11.0" N	117° 48' 1.0" W	0.8141	
E16	47.925278, -117.355833	47° 55' 31.0" N	117° 21' 21.0" W	0.8141	
E18	42.689167, -120.536389	42° 41' 21.0" N	120° 32' 11.0" W	0.6565	
E19	42.466944, -117.939167	42° 28' 1.0" N	117° 56' 21.0" W	0.6165	
E20	44.550278, -117.425278	44° 33' 1.0" N	117° 25' 31.0" W	0.5078	
E21	47.508611, -121.189167	47° 30' 31.0" N	121° 11' 21.0" W	0.4785	
E22	43.647500, -120.189167	43° 38' 51.0" N	120° 11' 21.0" W	0.4784	
E23	43.286389, -117.605833	43° 17' 11.0" N	117° 36' 21.0" W	0.4784	

E25	44.814167, -117.841944	44° 48' 51.0" N	117° 50' 31.0" W	0.4410	
E26	46.911389, -120.328056	46° 54' 41.0" N	120° 19' 41.0" W	0.3705	
E27	43.091944, -117.091944	43° 5' 31.0" N	117° 5' 31.0" W	0.3673	
E29	42.161389, -120.453056	42° 9' 41.0" N	120° 27' 11.0" W	0.2929	
E30	46.078056, -118.925278	46° 4' 41.0" N	118° 55' 31.0" W	0.2774	
E31	46.244722, -118.008611	46° 14' 41.0" N	118° 0' 31.0" W	0.2714	
E32	46.383611, -117.189167	46° 23' 1.0" N	117° 11' 21.0" W	0.2524	
E33	46.536389, -120.800278	46° 32' 11.0" N	120° 48' 1.0" W	0.2510	
E34	42.661389, -120.425278	42° 39' 41.0" N	120° 25' 31.0" W	0.2300	
E35	46.911389, -118.550278	46° 54' 41.0" N	118° 33' 1.0" W	0.1345	
E38	42.175278, -120.550278	42° 10' 31.0" N	120° 33' 1.0" W	0.0664	
E40	43.855833, -120.841944	43° 51' 21.0" N	120° 50' 31.0" W	0.0324	
E42	43.703056, -117.091944	43° 42' 11.0" N	117° 5' 31.0" W	0.0000	

Project IceWave | Pleistocene Megafauna Locality Intelligence v3 | West AUC 0.89 ★ | East AUC 0.846 ◆ | E01 Blind Validation ✓ | Data: PBDB (5 states), iDigBio, USGS 3DEP, USGS SGMC | github.com/bdgroves/project_ice_wave | March 2026